

Summary

Data Scientist with experience in machine learning, data mining, and analytics-driven decision making. Specializing in predictive modeling, data quality engineering, and large-scale data analysis. Proficient in Python and SQL for building end-to-end data pipelines, developing classification and regression models, and extracting actionable insights from structured and unstructured data. Experienced in applying advanced techniques including Random Forest, Gradient Boosting, XGBoost, NN, CNN, and RNN to real-world datasets. Skilled in data preprocessing, feature engineering, and model evaluation, with a strong focus on translating complex data into business impact through dashboards, reporting, and stakeholder collaboration.

Tools Utilized

Python | R | MySQL | Java | C++ | PowerShell | FastAPI | React | Docker | HuggingFace Transformers | RoBERTa | Grad-CAM | PySpark | Microsoft Fabric | Data Lakehouse | Data Warehouse | Captum | Spring Boot | RESTful APIs | PyTorch | pandas | NumPy | PySpark | scikit-learn | Matplotlib | Seaborn | ggplot2 | Power BI | Microsoft Excel | Office 365 | Jupyter Notebook | RStudio | Google Colab | Postman | Git

Education

Towson University | Master of Science - Computer Science - Data Science
Texas A&M University | Bachelor of Science - Applied Mathematics

Work Experience**Data Science Intern - 06/01/2026-Current**

CPower (an NRG Company) - Baltimore, MD

- Ingested hourly utility meter data across 1,200 CAISO sites over a 6-month period (~5M+ readings) from an internal API using PySpark in Microsoft Fabric, structuring it into a Delta Lake table.
- Applied statistical data quality scoring methods to evaluate hourly meter readings at scale, detecting outliers, flatlines, zeros, missing data, and nulls; assigned an overall account health grade and monthly grade per account, writing results to the warehouse for downstream use.
- Developed a full-stack analytics dashboard using FastAPI backend and React frontend to display data the quality grades aggregated by month, company, and account, used by the Enrollment Settlements and Payments team to support customer payout decisions.
- Engineered time-based and lag features from hourly time-series data and trained a multi-output XGBoost model to predict kW usage for 200 CAISO Target Corporations, achieving an R^2 of 0.93.
- Combined the ML predictions with my rule-based logic to build an anomaly detection system for identifying irregularities in customer load curtailment behavior, supporting the ESP team with more accurate data and faster customer payout processing.

Graduate Teaching Assistant, - 01/12/2026 to 05/12/2026

Towson University - Department of Computer Science

- Assisted in teaching Data Structures and Object-Oriented Programming (Java), guiding students through algorithmic problem solving and computational thinking.
- Evaluated and debugged student code, identifying logical errors and improving program efficiency and correctness.
- Supported lab sessions and provided technical explanations of core concepts including data structures and algorithm design.

Business Intelligence Developer Intern - 06/01/2025 to 08/14/2025

CPower - Baltimore, MD

- Developed Python-based data quality scripts to analyze and validate CRM data (Microsoft Dynamics 365), identifying ~8.5% of leads as low-quality ("Gray/Bad") and improving overall data reliability.
- Implemented fuzzy matching algorithms to detect duplicate accounts across Dataverse and SharePoint, identifying ~250 duplicate records and improving data integrity.
- Performed data analysis to uncover patterns in customer and operational data, supporting business decisions and prioritizing ~75 high-value national customers.

Projects**Humor Detection with Explainable AI (Python, PyTorch, Hugging Face, RoBERTa, FastAPI, React, Captum) [GitHub](#)**

- Built and fine-tuned a RoBERTa-based humor classification model using PyTorch and Hugging Face Transformers, developing preprocessing and tokenization pipelines to transform raw user text into model-ready inputs for accurate real-time predictions
- Applied Captum Integrated Gradients to analyze token-level feature importance, improving model transparency by identifying which text tokens contributed most to classification results.
- Improved model performance by fine-tuning, increasing accuracy from 56% to 98% and improving the humor detection F1-score from 0.22 to 0.98.
- Developed a full-stack machine learning application using React and FastAPI, allowing users to submit custom text inputs, receive real-time humor predictions, and visualize Integrated Gradients token-level attribution scores explaining model decisions.

Brain Tumor Classification with Explainable AI (Python, PyTorch, CNNs, Transfer Learning, Grad-CAM) [GitHub](#)

- Built and trained a custom convolutional neural network (CNN) for brain tumor classification using PyTorch, implementing image preprocessing pipelines to transform MRI scans into inputs for tumor detection.
- Improved model performance by applying transfer learning with DenseNet-121, increasing accuracy from 79% to 88% and improving meningioma tumor F1-score from 0.58 to 0.77 compared to the custom CNN baseline.
- Applied Grad-CAM Explainable AI techniques to generate class activation heatmaps, visualizing MRI regions that contributed most to tumor classification decisions and improving model interpretability.

CNN-Based Computer Vision System for Dog Breed Classification and Human-Dog Mapping (Python, PyTorch) [GitHub](#)

- Developed a computer vision application that detects whether an image contains a human or a dog, classifies dog breeds across 133 categories, and maps human faces to visually similar dog breeds. Improved model accuracy from ~13% to ~86% by replacing a baseline CNN with DenseNet-161 transfer learning.
- Enabled interactive and user-facing functionality by supporting real-time image classification and human-to-breed matching across a large multi-class dataset.

Subreddit Sentiment Classifier (roBERTa, Python)

- Fine-tuned a RoBERTa transformer model to classify Reddit comments into quality categories (Controversial, Baseline, High-Quality, Viral).
- Developed an NLP workflow including preprocessing, transformer tokenization, and supervised training, achieving $F1 \approx 0.79$ on validation data.

Natural Language to SQL – LLM Prompt Engineering (Python, OpenAI API)

- Developed a Python solution to measure how reliable LLMs translate natural language into executable SQL queries through OpenAI API.
- Ran controlled prompt-engineering experiments (zero-shot, few-shot, chain-of-thought) to quantify accuracy tradeoffs, identify failures, and determine strategies that improve reliability of LLM-generated queries.

MovieLens Recommender System (Python) [GitHub](#)

- Built collaborative filtering and ALS matrix factorization recommenders to generate personalized movie suggestions from user behavior data.
- Quantified improvements using ranking metrics (Precision@K, Recall@K, MAP, NDCG), showing superior recommendation relevance compared to baseline similarity models.
- Demonstrated how latent-factor recommendation systems drive user engagement and retention through personalized content discovery.